GoFixCarz
Admin-Only API Requirements

DRAFT FOR CLIENT APPROVAL

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Based on GoFixCarz Admin User Stories: 29 stories across 13 epics.

Admin-only contracts with request payloads, sample responses, permission rules, reporting formulas and client approval decisions.

Contents

1. Scope, access and conventions

2. Admin operation index

3. Detailed admin contracts

4. Existing baseline and requirements gaps

5. Story-to-API traceability

6. Client decision register

1. Scope, access and conventions

Proposed business API base path: /api/v1/admin. Relative paths below are admin-only and require an authenticated authorized admin. Authentication URL locations are never printed; the private operation entries cannot be used as URL requests.

Authorization: Bearer <admin_access_token>. Validate active account, session and effective permission/scope on every operation. Garage/customer tokens are not valid substitutes. A global CEO scope is assigned explicitly, not inferred from signup.

GET has no request body; required opaque path IDs use readable synthetic values here (UUID recommended). Query booleans/numbers are typed. Reject unsupported fields, negative pagination and invalid enum/filter values.

Pagination: page >= 1, default page_size=20, maximum 100; cursor message pagination default limit=30, maximum 100. Empty lists return items: []. Path IDs and ownership/scope are validated on each request.

Date filters are inclusive local dates in timezone (default proposed Asia/Kolkata). Implement as start inclusive/end exclusive after timezone conversion. date_from <= date_to; set maximum ranges for expensive reports. Returned filters and generated_at make the actual window explicit.

Timestamp: ISO 8601 with timezone. Money: integer paise with currency INR; 118000 = INR 1,180. Examples have invoice subtotal 100000 + GST 18000 = 118000. Currency conversions and profit calculations are not introduced.

Common success envelope:

```
{
  "success": true,
  "message": "Dashboard retrieved",
  "data": {
    "snapshot_id": "snapshot_001"
  }
}
```

Common error envelope:

```
{
  "success": false,
  "message": "You do not have permission to suspend garages",
  "error": {
    "code": "PERMISSION_DENIED",
    "details": {
      "required_permission": "garages.status.write"
    }
  }
}
```

Common failures: 401 ADMIN_AUTH_REQUIRED / SESSION_EXPIRED; 403 PERMISSION_DENIED / DATA_SCOPE_DENIED; 404 RESOURCE_NOT_FOUND; 422 VALIDATION_ERROR; 429 RATE_LIMITED. Endpoint-specific failures use the same envelope. Hide outside-scope record existence where appropriate.

Mutation idempotency and optimistic version checks protect duplicate sends/exports and conflicting edits. Required reason fields are nonblank strings, proposed maximum 500 characters. Audit all privileged changes; never log secrets.

Static routes such as /jobs/delays and /bookings/exception-statistics must be registered ahead of dynamic ID routes or otherwise explicitly disambiguated in implementation.

Priority remains the source Must Have / Should Have; operational scope is limited to the stories. The sign-off does not authorize broad new writes such as refund initiation, customer blocking, job editing or admin self-registration.

2. Admin operation index

ID	Method / proposed admin path or private operation	Permission
ADM-01	POST Private operation: begin admin sign-in	Pre-session admin verification
ADM-02	POST Private operation: complete admin MFA	MFA challenge
ADM-03	POST Private operation: refresh admin session	Refresh credential
ADM-04	POST Private operation: revoke current session	Authenticated admin session
ADM-05	GET /me	Authenticated admin
ADM-06	GET /menus	Authenticated admin
ADM-07	GET /permissions	security.read
ADM-08	GET /roles	security.read
ADM-09	GET /staff	security.read
ADM-10	PUT /staff/{admin_id}/roles	security.manage
ADM-11	GET /dashboard	dashboard.read
ADM-12	GET /analytics	dashboard.read
ADM-13	GET /garage-owners	garages.read
ADM-14	GET /garage-owners/{owner_id}	garages.read
ADM-15	GET /garages	garages.read
ADM-16	GET /garages/{garage_id}	garages.read
ADM-17	PATCH /garages/{garage_id}/status	garages.status.write
ADM-18	PATCH /garages/{garage_id}/verification	garages.verify
ADM-19	GET /garages/{garage_id}/reviews	garages.read
ADM-20	GET /customers	customers.read
ADM-21	GET /customers/{customer_id}	customers.read
ADM-22	GET /customers/{customer_id}/vehicles	customers.read
ADM-23	GET /customers/{customer_id}/reviews	customers.read
ADM-24	GET /bookings	bookings.read
ADM-25	GET /bookings/{booking_id}	bookings.read
ADM-26	GET /bookings/exception-statistics	bookings.read
ADM-27	GET /jobs	jobs.read
ADM-28	GET /jobs/{job_id}	jobs.read
ADM-29	GET /jobs/{job_id}/timeline	jobs.read
ADM-30	GET /jobs/delays	jobs.read
ADM-31	GET /alerts	jobs.read
ADM-32	GET /finance/summary	finance.read
ADM-33	GET /finance/trends	finance.read
ADM-34	GET /invoices	finance.read
ADM-35	GET /invoices/{invoice_id}	finance.read
ADM-36	GET /payments	finance.read
ADM-37	GET /payments/{payment_id}	finance.read
ADM-38	GET /invoices/{invoice_id}/payment-attempts	finance.read
ADM-39	GET /refunds	finance.read
ADM-40	GET /disputes	finance.read
ADM-41	GET /disputes/{dispute_id}	finance.read
ADM-42	GET /notifications	notifications.read
ADM-43	GET /notifications/{notification_id}	notifications.read
ADM-44	POST /announcements/preview	announcements.send
ADM-45	POST /announcements	announcements.send
ADM-46	GET /announcements	notifications.read
ADM-47	GET /announcements/{announcement_id}	notifications.read
ADM-48	GET /announcements/{announcement_id}/deliveries	notifications.read
ADM-49	GET /support/tickets	support.read
ADM-50	GET /support/tickets/{ticket_id}	support.read
ADM-51	PATCH /support/tickets/{ticket_id}/assignment	support.assign
ADM-52	PATCH /support/tickets/{ticket_id}/priority	support.manage
ADM-53	GET /support/tickets/{ticket_id}/messages	support.read
ADM-54	POST /support/tickets/{ticket_id}/messages	support.reply
ADM-55	PATCH /support/tickets/{ticket_id}/status	support.manage
ADM-56	GET /garage-owner-statistics	owner_statistics.read + finance.read for monetary fields
ADM-57	GET /garage-owner-statistics/summary	owner_statistics.read + finance.read for monetary fields
ADM-58	GET /garage-owner-statistics/{owner_id}	owner_statistics.read
ADM-59	GET /garage-owner-statistics/{owner_id}/trends	owner_statistics.read
ADM-60	POST /reports/exports	reports.export + selected report read permissions
ADM-61	GET /reports/exports/{export_id}	reports.export + selected report read permissions

ID	Method / proposed admin path or private operation	Permission
ADM-62	GET /reports/exports/{export_id}/download	reports.export + selected report read permissions
ADM-63	GET /audit-logs	audit.read
ADM-64	GET /audit-logs/{audit_id}	audit.read
ADM-65	GET /activity	activity.read
ADM-66	GET /activity/service-funnels/{booking_id}	activity.read + relevant linked-record permissions

3. Detailed admin contracts

Secure admin access - private route locations withheld

Stories: AUS-1

• The CSV requires secure login with MFA/OTP support and secure expiry. Shared documents must not reveal the private admin sign-in location. Logical operations below intentionally have no URL; location and security configuration are distributed privately.

• Admin accounts are separate from garage-owner/customer identities. No public registration or self-promotion is permitted. Passwords and codes are placeholders; never include real credentials in client documents.

• Proposed baseline: password plus mandatory MFA for administrative access. TOTP is recommended; an approved verified OTP factor may be used instead. Challenge and proof must be purpose-bound, short-lived, single-use and rate-limited.

• Invalid credentials use generic errors. Token rotation/reuse detection, secure device storage, session revocation, recent MFA for privileged actions and administrative session timeout need final approval. Existing password/session support does not imply MFA is implemented.

ADM-01 | Private operation: begin admin sign-in

POST Private authentication URL withheld

Permission / proof: Pre-session admin verification | Success HTTP: 200

Field validation / operation rules: identifier and password required; device_name optional. Do not issue a business-data access token before the required factor passes.

Request payload:

```
{
  "identifier": "admin@example.test",
  "password": "<password>",
  "device_name": "Admin workstation"
}
```

Sample success response:

```
{
  "success": true,
  "message": "Private operation: begin admin sign-in successful",
  "data": {
    "challenge_id": "admin_challenge_001",
    "mfa_required": true,
    "mfa_method": "TOTP",
    "expires_in_seconds": 300
  }
}
```

Representative failures: 401 INVALID_CREDENTIALS; 429 RATE_LIMITED. Common authorization and validation failures also apply.

ADM-02 | Private operation: complete admin MFA

POST Private authentication URL withheld

Permission / proof: MFA challenge | Success HTTP: 200

Field validation / operation rules: Both fields required. code is a string. This is a proposed private MFA contract, not an exposed route.

Request payload:

```
{
  "challenge_id": "admin_challenge_001",
  "code": "123456"
}
```

Sample success response:

```
{
  "success": true,
  "message": "Private operation: complete admin MFA successful",
  "data": {
    "admin": {
      "id": "admin_001",
      "display_name": "Example Administrator",
      "role_ids": [
        "role_ceo"
      ]
    },
    "access_token": "<admin_access_token>",
    "refresh_token": "<admin_refresh_token>",
    "expires_in_seconds": 900,
    "session_id": "admin_session_001"
  }
}
```

Representative failures: 400 MFA_INVALID; 400 MFA_EXPIRED; 429 RATE_LIMITED. Common authorization and validation failures also apply.

ADM-03 | Private operation: refresh admin session

POST Private authentication URL withheld

Permission / proof: Refresh credential | Success HTTP: 200

Field validation / operation rules: refresh_token required; reject revoked/expired/reused credentials.

Request payload:

```
{
  "refresh_token": "<admin_refresh_token>"
}
```

Sample success response:

```
{
  "success": true,
  "message": "Private operation: refresh admin session successful",
  "data": {
    "access_token": "<new_admin_access_token>",
    "refresh_token": "<rotated_admin_refresh_token>",
    "expires_in_seconds": 900
  }
}
```

Representative failures: 401 SESSION_EXPIRED; 401 REFRESH_REUSE_DETECTED. Common authorization and validation failures also apply.

ADM-04 | Private operation: revoke current session

POST Private authentication URL withheld

Permission / proof: Authenticated admin session | Success HTTP: 200

Field validation / operation rules: Revoke current session and refresh credentials; protected access must honor revocation.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Private operation: revoke current session successful",
  "data": {
    "logged_out": true
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

Identity, roles and permission-aware navigation

Stories: AUS-20

• RBAC is enforced server-side for every read, export and mutation, not only by hiding screens. Cross-garage access is limited to the admin's permitted data scope.

• Proposed functional permissions are shown per operation. CEO may receive approved full access; operations, finance and support roles receive least privilege. Role names do not bypass permission checks.

• Role assignment requires security.manage and recent MFA. Prevent self-escalation and removing the last active security administrator. Log before/after permission changes and revoke/re-evaluate affected sessions.

• Staff/role reads and role assignment are required to support AUS-20. Creating/deleting staff accounts or dynamic role creation is outside these stories; provisioning stays deliberate and private.

ADM-05 | Read admin identity and effective permissions

GET /api/v1/admin/me

Permission / proof: Authenticated admin | Success HTTP: 200

Field validation / operation rules: Return effective permissions, permitted garage scope and current identity; never return credential hashes.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Read admin identity and effective permissions successful",
  "data": {
    "id": "admin_001",
    "display_name": "Example Administrator",
    "role_ids": [
      "role_ceo"
    ],
    "permissions": [
      "dashboard.read",
      "garages.read",
      "finance.read",
      "reports.export"
    ],
    "data_scope": {
      "type": "PLATFORM"
    }
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-06 | Read authorized admin menus

GET /api/v1/admin/menus

Permission / proof: Authenticated admin | Success HTTP: 200

Field validation / operation rules: Return only authorized menu keys; private sign-in routes are never included.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Read authorized admin menus successful",
  "data": {
    "items": [
      {
        "key": "dashboard",
        "label": "Dashboard",
        "permission": "dashboard.read"
      },
      {
        "key": "finance",
        "label": "Finance",
        "permission": "finance.read"
      }
    ]
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-07 | Read permission catalogue

GET /api/v1/admin/permissions

Permission / proof: security.read | Success HTTP: 200

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Read permission catalogue successful",
  "data": {
    "items": [
      {
        "code": "garages.status.write",
        "description": "Activate or suspend a garage"
      },
      {
        "code": "security.manage",
        "description": "Assign admin staff roles"
      }
    ]
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-08 | Read approved admin roles

GET /api/v1/admin/roles

Permission / proof: security.read | Success HTTP: 200

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Read approved admin roles successful",
  "data": {
    "items": [
      {
        "id": "role_operations",
        "name": "Operations",
        "permissions": [
          "garages.read",
          "bookings.read",
          "jobs.read"
        ]
      },
      {
        "id": "role_finance",
        "name": "Finance",
        "permissions": [
          "finance.read",
          "reports.export"
        ]
      }
    ]
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-09 | List admin staff and assigned roles

GET /api/v1/admin/staff

Permission / proof: security.read | Success HTTP: 200

Sample query: ?search=Operations&is_active=true&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "List admin staff and assigned roles successful",
  "data": {
    "items": [
      {
        "id": "admin_002",
        "display_name": "Example Operations Staff",
        "is_active": true,
        "role_ids": [
          "role_operations"
        ]
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-10 | Assign approved roles/data scope

PUT /api/v1/admin/staff/{admin_id}/roles

Permission / proof: security.manage | Success HTTP: 200

Field validation / operation rules: All fields required. role_ids must be valid approved roles; type is PLATFORM or GARAGES. GARAGES requires nonempty unique garage_ids. Reject escalation outside grantor authority.

Request payload:

```
{
  "role_ids": [
    "role_operations"
  ],
  "data_scope": {
    "type": "GARAGES",
    "garage_ids": [
      "garage_001"
    ]
  },
  "reason": "Regional operations assignment"
}
```

Sample success response:

```
{
  "success": true,
  "message": "Assign approved roles/data scope successful",
  "data": {
    "admin_id": "admin_002",
    "role_ids": [
      "role_operations"
    ],
    "data_scope": {
      "type": "GARAGES",
      "garage_ids": [
        "garage_001"
      ]
    },
    "audit_id": "audit_001"
  }
}
```

Representative failures: 403 PRIVILEGE_ESCALATION_DENIED; 409 LAST_SECURITY_ADMIN; 422 ROLE_INVALID. Common authorization and validation failures also apply.

Executive dashboard and consistent analytics

Stories: AUS-2, AUS-3

• Common filters: date_from, date_to, timezone, city, garage_id, service_id. Apply the same normalized filters, permission scope and snapshot to all cards/charts.

• Server generates snapshot_id/as_of on initial read. Clients may pass snapshot_id on subsequent related dashboard/statistics/export reads. Expired snapshots produce 409 and require reload.

• Dashboard customer/garage counts are distinct participants in the selected segment/time range; lifetime totals require a separate explicitly-labelled view. Active job statuses are OPEN, IN_PROGRESS, WAITING_FOR_PARTS, QUALITY_CHECK and READY.

• Financial net_collected_amount means confirmed gross captured payments minus settled refunds, including tax; it is not profit or statutory revenue. Return taxes/discounts separately and approve the accounting definition.

ADM-11 | Read executive KPI dashboard

GET /api/v1/admin/dashboard

Permission / proof: dashboard.read | Success HTTP: 200

Sample query: ?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad

Field validation / operation rules: Filters are optional; default date window must be returned explicitly. Proposed metrics extend the existing dashboard contract.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Read executive KPI dashboard successful",
  "data": {
    "filters": {
      "date_from": "2026-10-01",
      "date_to": "2026-10-08",
      "timezone": "Asia/Kolkata",
      "city": "Hyderabad",
      "garage_id": null,
      "service_id": null
    },
    "snapshot_id": "snapshot_001",
    "generated_at": "2026-10-08T15:00:00+05:30",
    "kpis": {
      "garages": 2,
      "customers": 40,
      "bookings": 60,
      "active_jobs": 17,
      "completed_jobs": 30,
      "cancelled_jobs": 3,
      "cancelled_bookings": 4,
      "payment_failures": 3,
      "gross_collected_amount": 3540000,
      "refunded_amount": 118000,
      "net_collected_amount": 3422000,
      "currency": "INR"
    }
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-12 | Read filtered time-series charts

GET /api/v1/admin/analytics

Permission / proof: dashboard.read | Success HTTP: 200

Sample query: ?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&period=daily

Field validation / operation rules: period supports daily, weekly, monthly or custom. Chart time buckets and metric definitions must match the dashboard.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Read filtered time-series charts successful",
  "data": {
    "filters": {
      "date_from": "2026-10-01",
      "date_to": "2026-10-08",
      "timezone": "Asia/Kolkata",
      "city": "Hyderabad",
      "garage_id": null,
      "service_id": null
    },
    "snapshot_id": "snapshot_001",
    "generated_at": "2026-10-08T15:00:00+05:30",
    "period": "daily",
    "series": [
      {
        "date": "2026-10-08",
        "bookings": 8,
        "jobs_created": 5,
        "jobs_completed": 3,
        "captured_amount": 354000,
        "refunded_amount": 0,
        "net_collected_amount": 354000
      }
    ],
    "currency": "INR"
  }
}
```


Representative failures: 422 DATE_RANGE_INVALID; 409 SNAPSHOT_EXPIRED. Common authorization and validation failures also apply.

Garage owners, profiles and operational controls

Stories: AUS-4, AUS-5, AUS-6

• Owner identity and garage identity are distinct; one owner can manage several garages. Lists include ratings, booking/job/revenue aggregates and selected filters.

• Operational status (ACTIVE, SUSPENDED) and verification status (UNVERIFIED, VERIFIED) are separate controls. Verification does not silently activate a suspended garage.

• Suspension blocks discovery/new bookings according to approved policy, while preserving historical records and handling existing work explicitly. Status changes are audited with reason, actor and timestamp.

• Support/complaint counts link to support tickets; garage profile includes active services, hours, invoices, reviews and recent activity without exposing credential or bank secrets.

ADM-13 | List garage owners and activity

GET /api/v1/admin/garage-owners

Permission / proof: garages.read | Success HTTP: 200

Sample query: ?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&search=Example&status=ACTIVE&min_rating=4&page=1&page_size=20

Field validation / operation rules: Optional search, owner status, city, min_rating, garage_id and reporting filters. Account status is not garage operational status.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "List garage owners and activity successful",
  "data": {
    "items": [
      {
        "id": "owner_001",
        "display_name": "Example Garage Owner",
        "garage_ids": [
          "garage_001"
        ],
        "city": "Hyderabad",
        "account_status": "ACTIVE",
        "average_rating": 4.6,
        "bookings": 20,
        "jobs_created": 30,
        "net_collected_amount": 2242000,
        "currency": "INR"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-14 | View owner and managed garages

GET /api/v1/admin/garage-owners/{owner_id}

Permission / proof: garages.read | Success HTTP: 200

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "View owner and managed garages successful",
  "data": {
    "id": "owner_001",
    "display_name": "Example Garage Owner",
    "account_status": "ACTIVE",
    "garages": [
      {
        "id": "garage_001",
        "name": "Example Garage",
        "operational_status": "ACTIVE",
        "verification_status": "VERIFIED"
      }
    ]
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-15 | List admin-visible garages

GET /api/v1/admin/garages

Permission / proof: garages.read | Success HTTP: 200

Sample query: ?city=Hyderabad&operational_status=ACTIVE&verification_status=VERIFIED&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "List admin-visible garages successful",
  "data": {
    "items": [
      {
        "id": "garage_001",
        "owner_id": "owner_001",
        "name": "Example Garage",
        "city": "Hyderabad",
        "operational_status": "ACTIVE",
        "verification_status": "VERIFIED",
        "average_rating": 4.6
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-16 | Inspect garage profile and operations

GET /api/v1/admin/garages/{garage_id}

Permission / proof: garages.read | Success HTTP: 200

Field validation / operation rules: Collections in this summary are bounded previews. Related bookings/jobs/invoices/support APIs allow garage_id filtering and pagination.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Inspect garage profile and operations successful",
  "data": {
    "id": "garage_001",
    "owner_id": "owner_001",
    "name": "Example Garage",
    "city": "Hyderabad",
    "timezone": "Asia/Kolkata",
    "operational_status": "ACTIVE",
    "verification_status": "VERIFIED",
    "services": [
      {
        "id": "service_001",
        "name": "Basic Service",
        "price_amount": 100000,
        "currency": "INR"
      }
    ],
    "working_hours": [
      {
        "day": "THURSDAY",
        "opens": "09:00",
        "closes": "18:00"
      }
    ],
    "summary": {
      "bookings": 20,
      "jobs": 30,
      "invoices": 20,
      "average_rating": 4.6,
      "complaints": 2
    },
    "recent_activity": [
      {
        "type": "JOB_CREATED",
        "job_id": "job_001",
        "timestamp": "2026-10-08T10:00:00+05:30"
      }
    ]
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-17 | Activate or suspend garage

PATCH /api/v1/admin/garages/{garage_id}/status

Permission / proof: garages.status.write | Success HTTP: 200

Field validation / operation rules: All fields required. Use expected_version for concurrency; sample change targets garage_002, separate from active garage_001.

Request payload:

```
{
  "operational_status": "SUSPENDED",
  "reason": "Partner quality review",
  "expected_version": 3
}
```

Sample success response:

```
{
  "success": true,
  "message": "Activate or suspend garage successful",
  "data": {
    "id": "garage_002",
    "operational_status": "SUSPENDED",
    "version": 4,
    "audit_id": "audit_002"
  }
}
```

Representative failures: 409 VERSION_CONFLICT; 422 INVALID_STATUS_TRANSITION. Common authorization and validation failures also apply.

ADM-18 | Verify garage

PATCH /api/v1/admin/garages/{garage_id}/verification

Permission / proof: garages.verify | Success HTTP: 200

Field validation / operation rules: All fields required. Verification/reversal transitions need policy; changing verification does not override operational suspension.

Request payload:

```
{
  "verification_status": "VERIFIED",
  "reason": "Required documents reviewed",
  "expected_version": 4
}
```

Sample success response:

```
{
  "success": true,
  "message": "Verify garage successful",
  "data": {
    "id": "garage_002",
    "operational_status": "SUSPENDED",
    "verification_status": "VERIFIED",
    "version": 5,
    "audit_id": "audit_003"
  }
}
```

Representative failures: 409 VERSION_CONFLICT; 422 VERIFICATION_REQUIREMENTS_NOT_MET. Common authorization and validation failures also apply.

ADM-19 | Inspect garage ratings/reviews

GET /api/v1/admin/garages/{garage_id}/reviews

Permission / proof: garages.read | Success HTTP: 200

Sample query: ?moderation_status=PUBLISHED&page=1&page_size=20

Field validation / operation rules: Inspection only; review moderation writes are not requested in this source.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Inspect garage ratings/reviews successful",
  "data": {
    "items": [
      {
        "id": "review_001",
        "customer_id": "customer_001",
        "job_id": "job_001",
        "rating": 4,
        "comment": "Good service",
        "moderation_status": "PUBLISHED"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

Customer account and risk monitoring

Stories: AUS-7, AUS-8

• Customer login identities must be distinguished from garage-owned contact records. Cross-platform customer reads require customers.read; do not mistake owner accounts for customer identities.

• Customer list filters include search, account_status, inactive_since, last activity, failed-payment count and support flags. Define the inactivity threshold in configuration.

• Profile shows paginated linked vehicles, bookings, jobs, invoices, payment attempts, reviews and support tickets. Return permission-filtered summaries and use related list endpoints for drill-down.

• The source requests identifying inactive/blocked accounts, not authority to block/delete customers. No customer-blocking, impersonation or credential reset operation is added.

ADM-20 | List customers and risk flags

GET /api/v1/admin/customers

Permission / proof: customers.read | Success HTTP: 200

Sample query: ?account_status=ACTIVE&has_failed_payments=true&has_support_flags=true&page=1&page_size=20

Field validation / operation rules: All filters optional and combined with AND. Optional inactive_since uses last_activity_at before the supplied cutoff, with never-active handling defined in policy.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "List customers and risk flags successful",
  "data": {
    "items": [
      {
        "id": "customer_001",
        "display_name": "Example Customer",
        "account_status": "ACTIVE",
        "last_activity_at": "2026-10-08T12:00:00+05:30",
        "failed_payments": 1,
        "support_flags": [
          "OPEN_PAYMENT_TICKET"
        ]
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-21 | Inspect customer and linked records

GET /api/v1/admin/customers/{customer_id}

Permission / proof: customers.read | Success HTTP: 200

Field validation / operation rules: Linked finance details additionally require finance.read. Paginate full related data via the dedicated lists; vehicle and review list operations below complete the profile drill-down.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect customer and linked records successful",
  "data": {
    "id": "customer_001",
    "display_name": "Example Customer",
    "account_status": "ACTIVE",
    "vehicles": [
      {
        "id": "vehicle_001",
        "registration_number": "TS00AA0001",
        "model": "Swift"
      }
    ],
    "bookings": [
      {
        "id": "booking_001",
        "status": "CONVERTED"
      }
    ],
    "jobs": [
      {
        "id": "job_001",
        "status": "COMPLETED"
      }
    ],
    "invoices": [
      {
        "id": "invoice_001",
        "total_amount": 118000,
        "payment_status": "PAID"
      }
    ],
    "payments": [
      {
        "id": "payment_001",
        "status": "PAID"
      }
    ],
    "reviews": [
      {
        "id": "review_001",
        "rating": 4
      }
    ],
    "support_tickets": [
      {
        "id": "ticket_001",
        "status": "OPEN"
      }
    ],
    "related_preview_limit": 5
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-22 | Inspect all customer vehicles

GET /api/v1/admin/customers/{customer_id}/vehicles

Permission / proof: customers.read | Success HTTP: 200

Sample query: ?page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect all customer vehicles successful",
  "data": {
    "items": [
      {
        "id": "vehicle_001",
        "registration_number": "TS00AA0001",
        "brand": "Maruti Suzuki",
        "model": "Swift",
        "fuel_type": "PETROL"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-23 | Inspect customer review history

GET /api/v1/admin/customers/{customer_id}/reviews

Permission / proof: customers.read | Success HTTP: 200

Sample query: ?page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect customer review history successful",
  "data": {
    "items": [
      {
        "id": "review_001",
        "garage_id": "garage_001",
        "job_id": "job_001",
        "rating": 4,
        "moderation_status": "PUBLISHED"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

Booking monitoring and exception trends

Stories: AUS-9, AUS-10

• Read-only monitoring; admin does not accept/reject or convert bookings on behalf of a garage in this requirements baseline.

• Booking states: PENDING, ACCEPTED, REJECTED, CANCELLED, CONVERTED, EXPIRED. Include actor, timestamps, rejection/cancellation/expiry reasons and linked job where available.

• Exception charts break down rejected/cancelled/expired bookings by reason, owner, city and time using the same filter/snapshot convention. Missing reasons are explicitly categorized as UNSPECIFIED.

ADM-24 | Monitor bookings across permitted garages

GET /api/v1/admin/bookings

Permission / proof: bookings.read | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&status=CONVERTED&customer_id=customer_001&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Monitor bookings across permitted garages successful",
  "data": {
    "items": [
      {
        "id": "booking_001",
        "customer_id": "customer_001",
        "garage_id": "garage_001",
        "owner_id": "owner_001",
        "vehicle_id": "vehicle_001",
        "service_ids": [
          "service_001"
        ],
        "starts_at": "2026-10-08T10:00:00+05:30",
        "status": "CONVERTED",
        "created_at": "2026-10-07T15:00:00+05:30",
        "accepted_at": "2026-10-07T16:00:00+05:30",
        "job_id": "job_001"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-25 | Inspect booking history/decision reasons

GET /api/v1/admin/bookings/{booking_id}

Permission / proof: bookings.read | Success HTTP: 200

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect booking history/decision reasons successful",
  "data": {
    "id": "booking_002",
    "status": "REJECTED",
    "customer_id": "customer_002",
    "garage_id": "garage_002",
    "owner_id": "owner_002",
    "reason": "Requested slot unavailable",
    "decision_actor": {
      "role": "GARAGE_OWNER",
      "id": "owner_002"
    },
    "decided_at": "2026-10-08T11:00:00+05:30",
    "timeline": [
      {
        "status": "PENDING",
        "timestamp": "2026-10-08T10:50:00+05:30"
      },
      {
        "status": "REJECTED",
        "timestamp": "2026-10-08T11:00:00+05:30"
      }
    ]
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-26 | Read booking exception breakdowns

GET /api/v1/admin/bookings/exception-statistics

Permission / proof: bookings.read | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&group_by=owner,reason,day

Field validation / operation rules: group_by supports owner, garage, reason, city, day. Date basis is the exception transition timestamp.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Read booking exception breakdowns successful",
  "data": {
    "filters": {
      "date_from": "2026-10-01",
      "date_to": "2026-10-08",
      "timezone": "Asia/Kolkata",
      "city": "Hyderabad",
      "garage_id": null,
      "service_id": null
    },
    "snapshot_id": "snapshot_001",
    "generated_at": "2026-10-08T15:00:00+05:30",
    "totals": {
      "REJECTED": 6,
      "CANCELLED": 4,
      "EXPIRED": 2
    },
    "by_reason": [
      {
        "reason": "UNSPECIFIED",
        "count": 2
      },
      {
        "reason": "Requested slot unavailable",
        "count": 10
      }
    ],
    "by_owner": [
      {
        "owner_id": "owner_001",
        "count": 7
      },
      {
        "owner_id": "owner_002",
        "count": 5
      }
    ]
  }
}
```


Representative failures: 422 INVALID_GROUPING; 409 SNAPSHOT_EXPIRED. Common authorization and validation failures also apply.

Job monitoring, timeline and delay alerts

Stories: AUS-11, AUS-12

• Job states: OPEN, IN_PROGRESS, WAITING_FOR_PARTS, QUALITY_CHECK, READY, COMPLETED, CANCELLED. Return complete authorized detail without granting job-edit/status-change powers.

• Delay detection uses configured thresholds per service/state and recorded timestamps. State duration starts at entry into the current state; overall overdue compares an approved due_at against as_of.

• Jobs waiting for parts are filterable independently of being overdue. Alerts show measured duration, threshold and the current delay reason. Configuration ownership must be approved.

ADM-27 | Monitor job cards

GET /api/v1/admin/jobs

Permission / proof: jobs.read | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&status=COMPLETED&customer_id=customer_001&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Monitor job cards successful",
  "data": {
    "items": [
      {
        "id": "job_001",
        "job_number": "GFC-J-000001",
        "garage_id": "garage_001",
        "customer_id": "customer_001",
        "vehicle_id": "vehicle_001",
        "status": "COMPLETED",
        "created_at": "2026-10-08T10:00:00+05:30",
        "completed_at": "2026-10-08T12:00:00+05:30"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-28 | Inspect job findings and charges

GET /api/v1/admin/jobs/{job_id}

Permission / proof: jobs.read | Success HTTP: 200

Field validation / operation rules: Mask restricted notes/financial fields where the admin lacks their permissions.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect job findings and charges successful",
  "data": {
    "id": "job_001",
    "garage_id": "garage_001",
    "customer_id": "customer_001",
    "vehicle_id": "vehicle_001",
    "status": "COMPLETED",
    "services": [
      {
        "id": "service_001",
        "name": "Basic Service"
      }
    ],
    "parts": [
      {
        "name": "Engine oil",
        "quantity": 1
      }
    ],
    "labour": [
      {
        "description": "Oil change"
      }
    ],
    "inspection_findings": [
      "Routine maintenance completed"
    ],
    "created_at": "2026-10-08T10:00:00+05:30",
    "completed_at": "2026-10-08T12:00:00+05:30"
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-29 | Inspect chronological job events

GET /api/v1/admin/jobs/{job_id}/timeline

Permission / proof: jobs.read | Success HTTP: 200

Sample query: ?page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Inspect chronological job events successful",
  "data": {
    "items": [
      {
        "status": "OPEN",
        "timestamp": "2026-10-08T10:00:00+05:30",
        "actor_id": "owner_001"
      },
      {
        "status": "COMPLETED",
        "timestamp": "2026-10-08T12:00:00+05:30",
        "actor_id": "owner_001"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 2
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-30 | Identify delayed/waiting jobs

GET /api/v1/admin/jobs/delays

Permission / proof: jobs.read | Success HTTP: 200

Sample query: ?status=WAITING_FOR_PARTS&overdue=true&garage_id=garage_001&page=1&page_size=20

Field validation / operation rules: Return as_of in list metadata; sample durations measured at 2026-10-08T15:00:00+05:30.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Identify delayed/waiting jobs successful",
  "data": {
    "items": [
      {
        "id": "job_003",
        "garage_id": "garage_001",
        "status": "WAITING_FOR_PARTS",
        "state_entered_at": "2026-10-06T10:00:00+05:30",
        "measured_state_hours": 53,
        "threshold_hours": 24,
        "is_overdue": true,
        "reason": "Part not received"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-31 | Read operational delay alerts

GET /api/v1/admin/alerts

Permission / proof: jobs.read | Success HTTP: 200

Sample query: ?type=JOB_DELAY&severity=HIGH&page=1&page_size=20

Field validation / operation rules: Alert generation/deduplication must be defined; this read API does not imply a customer notification or intervention mutation.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Read operational delay alerts successful",
  "data": {
    "items": [
      {
        "id": "alert_001",
        "type": "JOB_DELAY",
        "severity": "HIGH",
        "job_id": "job_003",
        "garage_id": "garage_001",
        "threshold_hours": 24,
        "measured_hours": 53,
        "detected_at": "2026-10-08T15:00:00+05:30"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

Finance, invoices, payments, refunds and disputes

Stories: AUS-13, AUS-14

• Finance is read-only investigation/reporting in the source. No payment-status override, refund initiation, charge, invoice-edit or missing-bill repair endpoint is proposed.

• Amounts are integer paise, INR. Full/partial refunds are separate settled financial events, not failed payment attempts. Reconcile capture/refund amounts before reporting.

• Sanitize gateway responses: expose approved response codes, timestamps and transaction references only. Exclude secrets, signatures, card/bank details and raw unrestricted gateway payloads.

• Financial reports include totals/trends, status, garage breakdown, tax and discount totals. Different event date bases must be explicit: invoice issued_at; payment captured_at; refund settled_at; dispute opened_at.

ADM-32 | Read financial totals and breakdown

GET /api/v1/admin/finance/summary

Permission / proof: finance.read | Success HTTP: 200

Sample query: ?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad

Field validation / operation rules: Return event-date bases and matching filter metadata; payment trends available through finance analytics.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Read financial totals and breakdown successful",
  "data": {
    "filters": {
      "date_from": "2026-10-01",
      "date_to": "2026-10-08",
      "timezone": "Asia/Kolkata",
      "city": "Hyderabad",
      "garage_id": null,
      "service_id": null
    },
    "snapshot_id": "snapshot_001",
    "generated_at": "2026-10-08T15:00:00+05:30",
    "gross_collected_amount": 3540000,
    "refunded_amount": 118000,
    "net_collected_amount": 3422000,
    "issued_invoice_total_amount": 3540000,
    "tax_amount": 540000,
    "discount_amount": 0,
    "failed_attempts": 3,
    "currency": "INR",
    "by_garage": [
      {
        "garage_id": "garage_001",
        "net_collected_amount": 2242000
      },
      {
        "garage_id": "garage_002",
        "net_collected_amount": 1180000
      }
    ]
  }
}
```


Representative failures: 422 DATE_RANGE_INVALID; 409 SNAPSHOT_EXPIRED. Common authorization and validation failures also apply.

ADM-33 | Read finance trends

GET /api/v1/admin/finance/trends

Permission / proof: finance.read | Success HTTP: 200

Sample query: ?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&period=daily

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Read finance trends successful",
  "data": {
    "filters": {
      "date_from": "2026-10-01",
      "date_to": "2026-10-08",
      "timezone": "Asia/Kolkata",
      "city": "Hyderabad",
      "garage_id": null,
      "service_id": null
    },
    "snapshot_id": "snapshot_001",
    "generated_at": "2026-10-08T15:00:00+05:30",
    "currency": "INR",
    "series": [
      {
        "date": "2026-10-08",
        "captured_amount": 354000,
        "refunded_amount": 0,
        "net_collected_amount": 354000,
        "tax_amount": 54000,
        "discount_amount": 0,
        "failed_attempts": 1
      }
    ]
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-34 | List invoices with billing totals

GET /api/v1/admin/invoices

Permission / proof: finance.read | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&customer_id=customer_001&payment_status=PAID&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "List invoices with billing totals successful",
  "data": {
    "items": [
      {
        "id": "invoice_001",
        "job_id": "job_001",
        "customer_id": "customer_001",
        "garage_id": "garage_001",
        "subtotal_amount": 100000,
        "tax_amount": 18000,
        "discount_amount": 0,
        "total_amount": 118000,
        "payment_status": "PAID",
        "currency": "INR"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-35 | Inspect invoice and payment links

GET /api/v1/admin/invoices/{invoice_id}

Permission / proof: finance.read | Success HTTP: 200

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Inspect invoice and payment links successful",
  "data": {
    "id": "invoice_001",
    "job_id": "job_001",
    "customer_id": "customer_001",
    "garage_id": "garage_001",
    "line_items": [
      {
        "type": "SERVICE",
        "description": "Basic Service",
        "quantity": 1,
        "unit_amount": 100000,
        "line_total": 100000
      }
    ],
    "subtotal_amount": 100000,
    "taxes": [
      {
        "name": "GST",
        "rate_percent": 18,
        "amount": 18000
      }
    ],
    "discount_amount": 0,
    "total_amount": 118000,
    "currency": "INR",
    "payment_status": "PAID",
    "payment_ids": [
      "payment_001"
    ]
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-36 | Find failed/disputed or settled payments

GET /api/v1/admin/payments

Permission / proof: finance.read | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&status=PAID&disputed=false&page=1&page_size=20

Field validation / operation rules: Optional customer_id, invoice_id, garage_id, status and disputed filters. Failure and dispute are different dimensions.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Find failed/disputed or settled payments successful",
  "data": {
    "items": [
      {
        "id": "payment_001",
        "invoice_id": "invoice_001",
        "customer_id": "customer_001",
        "garage_id": "garage_001",
        "status": "PAID",
        "amount": 118000,
        "currency": "INR",
        "dispute_status": "NONE"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-37 | Inspect payment attempt and safe gateway result

GET /api/v1/admin/payments/{payment_id}

Permission / proof: finance.read | Success HTTP: 200

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect payment attempt and safe gateway result successful",
  "data": {
    "id": "payment_failed_001",
    "invoice_id": "invoice_002",
    "customer_id": "customer_002",
    "garage_id": "garage_002",
    "status": "FAILED",
    "amount": 118000,
    "currency": "INR",
    "gateway": {
      "provider": "<configured_provider>",
      "order_reference": "<synthetic_order_reference>",
      "response_code": "PAYMENT_DECLINED",
      "responded_at": "2026-10-08T12:00:00+05:30"
    },
    "dispute_ids": [],
    "related_attempt_ids": [
      "payment_failed_001"
    ]
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-38 | Inspect every attempt for invoice

GET /api/v1/admin/invoices/{invoice_id}/payment-attempts

Permission / proof: finance.read | Success HTTP: 200

Sample query: ?page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect every attempt for invoice successful",
  "data": {
    "items": [
      {
        "id": "payment_001",
        "status": "PAID",
        "amount": 118000,
        "created_at": "2026-10-08T12:15:00+05:30"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-39 | Inspect refund records

GET /api/v1/admin/refunds

Permission / proof: finance.read | Success HTTP: 200

Sample query: ?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&status=SETTLED&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect refund records successful",
  "data": {
    "items": [
      {
        "id": "refund_001",
        "payment_id": "payment_001",
        "invoice_id": "invoice_001",
        "garage_id": "garage_001",
        "amount": 11800,
        "currency": "INR",
        "status": "SETTLED",
        "settled_at": "2026-10-08T13:00:00+05:30"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-40 | Find disputed payments

GET /api/v1/admin/disputes

Permission / proof: finance.read | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&status=OPEN&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Find disputed payments successful",
  "data": {
    "items": [
      {
        "id": "dispute_001",
        "payment_id": "payment_003",
        "invoice_id": "invoice_003",
        "customer_id": "customer_003",
        "garage_id": "garage_001",
        "status": "OPEN",
        "reason": "Customer reports duplicate payment"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-41 | Inspect dispute context and evidence

GET /api/v1/admin/disputes/{dispute_id}

Permission / proof: finance.read | Success HTTP: 200

Field validation / operation rules: Read-only evidence; resolving/closing a gateway dispute needs separate scope.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect dispute context and evidence successful",
  "data": {
    "id": "dispute_001",
    "payment_id": "payment_003",
    "invoice_id": "invoice_003",
    "customer_id": "customer_003",
    "garage_id": "garage_001",
    "status": "OPEN",
    "reason": "Customer reports duplicate payment",
    "support_ticket_id": "ticket_002",
    "timeline": [
      {
        "event": "DISPUTE_OPENED",
        "timestamp": "2026-10-08T14:00:00+05:30"
      }
    ]
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

Notification monitoring and operational announcements

Stories: AUS-15, AUS-16

• Notifications are searchable by recipient role/ID, event type, booking/job/invoice, delivery status and timestamp. In-app read status and push delivery status are distinct.

• Announcements select a permitted customer/garage audience, store message, audience snapshot, actor, delivery outcomes and audit record. Preview before send, use idempotency and bound recipient counts.

• Sending operational announcements does not authorize marketing campaigns. Permissions, channel consent, text limits, delivery provider, recipient scope and suppression policies need approval.

ADM-42 | Inspect platform notification delivery

GET /api/v1/admin/notifications

Permission / proof: notifications.read | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&recipient_role=CUSTOMER&event_type=INVOICE_PUBLISHED&status=DELIVERED&page=1&page_size=20

Field validation / operation rules: Optional recipient_id, booking_id, job_id and invoice_id filters.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect platform notification delivery successful",
  "data": {
    "items": [
      {
        "id": "notification_001",
        "recipient": {
          "role": "CUSTOMER",
          "id": "customer_001"
        },
        "event_type": "INVOICE_PUBLISHED",
        "invoice_id": "invoice_001",
        "delivery_status": "DELIVERED",
        "read_at": null,
        "created_at": "2026-10-08T12:10:00+05:30"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-43 | Inspect safe delivery attempts

GET /api/v1/admin/notifications/{notification_id}

Permission / proof: notifications.read | Success HTTP: 200

Field validation / operation rules: Do not expose recipient device tokens, provider credentials or raw provider responses.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect safe delivery attempts successful",
  "data": {
    "id": "notification_001",
    "recipient": {
      "role": "CUSTOMER",
      "id": "customer_001"
    },
    "title": "Invoice available",
    "body": "Your final invoice is ready.",
    "attempts": [
      {
        "channel": "PUSH",
        "status": "DELIVERED",
        "attempted_at": "2026-10-08T12:10:00+05:30",
        "provider_code": "ACCEPTED"
      }
    ]
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-44 | Preview announcement audience

POST /api/v1/admin/announcements/preview

Permission / proof: announcements.send | Success HTTP: 200

Field validation / operation rules: All fields required. Resolve recipients within caller scope; proposed title limit 120, body limit 2,000.

Request payload:

```
{
  "audience": {
    "recipient_roles": [
      "GARAGE_OWNER"
    ],
    "city": "Hyderabad",
    "garage_ids": [
      "garage_001"
    ]
  },
  "channels": [
    "IN_APP",
    "PUSH"
  ],
  "title": "Operational update",
  "body": "Please review appointment availability."
}
```

Sample success response:

```
{
  "success": true,
  "message": "Preview announcement audience successful",
  "data": {
    "preview_id": "preview_001",
    "recipient_count": 1,
    "expires_at": "2026-10-08T15:10:00+05:30"
  }
}
```

Representative failures: 422 AUDIENCE_INVALID; 413 AUDIENCE_TOO_LARGE. Common authorization and validation failures also apply.

ADM-45 | Queue approved operational announcement

POST /api/v1/admin/announcements

Permission / proof: announcements.send | Success HTTP: 202

Required header: Idempotency-Key: <unique_key>

Field validation / operation rules: Both fields required; recent MFA recommended. Persist audience/message snapshot from preview; do not accept changed content with an old preview.

Request payload:

```
{
  "preview_id": "preview_001",
  "reason": "Appointment capacity review"
}
```

Sample success response:

```
{
  "success": true,
  "message": "Queue approved operational announcement successful",
  "data": {
    "id": "announcement_001",
    "status": "QUEUED",
    "recipient_count": 1,
    "audit_id": "audit_004"
  }
}
```

Representative failures: 409 PREVIEW_EXPIRED; 409 IDEMPOTENCY_CONFLICT. Common authorization and validation failures also apply.

ADM-46 | List sent/queued announcements

GET /api/v1/admin/announcements

Permission / proof: notifications.read | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&status=COMPLETED&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "List sent/queued announcements successful",
  "data": {
    "items": [
      {
        "id": "announcement_001",
        "title": "Operational update",
        "status": "COMPLETED",
        "recipient_count": 1,
        "delivered_count": 1,
        "failed_count": 0
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-47 | Inspect audience and delivery totals

GET /api/v1/admin/announcements/{announcement_id}

Permission / proof: notifications.read | Success HTTP: 200

Field validation / operation rules: Recipient totals are distinct recipients; per-channel delivery attempts are separately listed.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Inspect audience and delivery totals successful",
  "data": {
    "id": "announcement_001",
    "status": "COMPLETED",
    "audience": {
      "recipient_roles": [
        "GARAGE_OWNER"
      ],
      "garage_ids": [
        "garage_001"
      ]
    },
    "channels": [
      "IN_APP",
      "PUSH"
    ],
    "recipient_count": 1,
    "delivered_count": 1,
    "failed_count": 0,
    "actor_id": "admin_001",
    "audit_id": "audit_004"
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-48 | Inspect announcement delivery outcomes

GET /api/v1/admin/announcements/{announcement_id}/deliveries

Permission / proof: notifications.read | Success HTTP: 200

Sample query: ?status=DELIVERED&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect announcement delivery outcomes successful",
  "data": {
    "items": [
      {
        "recipient_id": "owner_001",
        "recipient_role": "GARAGE_OWNER",
        "channel": "PUSH",
        "status": "DELIVERED"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

Support ticket assignment, priority, replies and closure

Stories: AUS-17

• Support permissions independently control reading, assigning, setting priority, replying and changing status. Assignment targets active authorized support staff.

• Proposed statuses: OPEN, IN_PROGRESS, WAITING_FOR_CUSTOMER, RESOLVED, CLOSED. Priorities: LOW, NORMAL, HIGH, URGENT. Approve permitted transitions and reopening policy.

• Ticket links may include customer, garage, booking, job, invoice and payment. Access to linked financial detail still requires finance.read. Every mutation is audited.

• Replies are customer/garage-visible according to the ticket participant context. Internal notes must never leak into external replies; this baseline provides external reply operations only.

ADM-49 | Find support tickets

GET /api/v1/admin/support/tickets

Permission / proof: support.read | Success HTTP: 200

Sample query:
?status=OPEN&priority=NORMAL&customer_id=customer_001&garage_id=garage_001&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Find support tickets successful",
  "data": {
    "items": [
      {
        "id": "ticket_001",
        "subject": "Invoice question",
        "customer_id": "customer_001",
        "garage_id": "garage_001",
        "status": "OPEN",
        "priority": "NORMAL",
        "assignee_id": null
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-50 | Inspect support ticket and record links

GET /api/v1/admin/support/tickets/{ticket_id}

Permission / proof: support.read | Success HTTP: 200

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect support ticket and record links successful",
  "data": {
    "id": "ticket_001",
    "subject": "Invoice question",
    "description": "Please clarify the charges.",
    "customer_id": "customer_001",
    "garage_id": "garage_001",
    "booking_id": "booking_001",
    "job_id": "job_001",
    "invoice_id": "invoice_001",
    "status": "OPEN",
    "priority": "NORMAL",
    "assignee_id": null,
    "version": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-51 | Assign/unassign support staff

PATCH /api/v1/admin/support/tickets/{ticket_id}/assignment

Permission / proof: support.assign | Success HTTP: 200

Field validation / operation rules: All fields required; assignee_id may be null to unassign. Validate scope and staff eligibility.

Request payload:

```
{
  "assignee_id": "admin_support_001",
  "reason": "Finance support queue",
  "expected_version": 1
}
```

Sample success response:

```
{
  "success": true,
  "message": "Assign/unassign support staff successful",
  "data": {
    "id": "ticket_001",
    "assignee_id": "admin_support_001",
    "version": 2,
    "audit_id": "audit_005"
  }
}
```

Representative failures: 422 ASSIGNEE_INVALID; 409 VERSION_CONFLICT. Common authorization and validation failures also apply.

ADM-52 | Change ticket priority

PATCH /api/v1/admin/support/tickets/{ticket_id}/priority

Permission / proof: support.manage | Success HTTP: 200

Field validation / operation rules: All fields required.

Request payload:

```
{
  "priority": "HIGH",
  "reason": "Customer service blocked",
  "expected_version": 2
}
```

Sample success response:

```
{
  "success": true,
  "message": "Change ticket priority successful",
  "data": {
    "id": "ticket_001",
    "priority": "HIGH",
    "version": 3,
    "audit_id": "audit_006"
  }
}
```

Representative failures: 422 PRIORITY_INVALID; 409 VERSION_CONFLICT. Common authorization and validation failures also apply.

ADM-53 | Read ticket messages

GET /api/v1/admin/support/tickets/{ticket_id}/messages

Permission / proof: support.read | Success HTTP: 200

Sample query: ?limit=30

Field validation / operation rules: cursor optional; limit 1..100.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Read ticket messages successful",
  "data": {
    "items": [
      {
        "id": "support_message_001",
        "sender_role": "CUSTOMER",
        "text": "Please clarify the charges.",
        "created_at": "2026-10-08T12:30:00+05:30"
      }
    ],
    "next_cursor": null
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-54 | Reply to customer/garage

POST /api/v1/admin/support/tickets/{ticket_id}/messages

Permission / proof: support.reply | Success HTTP: 201

Field validation / operation rules: Both fields required; text nonblank, proposed maximum 2,000 characters. Deduplicate client_message_id.

Request payload:

```
{
  "client_message_id": "admin_reply_001",
  "text": "We are reviewing your invoice."
}
```

Sample success response:

```
{
  "success": true,
  "message": "Reply to customer/garage successful",
  "data": {
    "id": "support_message_002",
    "sender_role": "ADMIN",
    "text": "We are reviewing your invoice.",
    "created_at": "2026-10-08T15:00:00+05:30",
    "audit_id": "audit_007"
  }
}
```

Representative failures: 409 TICKET_CLOSED; 409 MESSAGE_ID_CONFLICT; 422 MESSAGE_INVALID. Common authorization and validation failures also apply.

ADM-55 | Resolve or close ticket

PATCH /api/v1/admin/support/tickets/{ticket_id}/status

Permission / proof: support.manage | Success HTTP: 200

Field validation / operation rules: status and expected_version required; resolution required for RESOLVED/CLOSED. Enforce approved transitions.

Request payload:

```
{
  "status": "CLOSED",
  "resolution": "Charges explained and confirmed by the customer.",
  "expected_version": 3
}
```

Sample success response:

```
{
  "success": true,
  "message": "Resolve or close ticket successful",
  "data": {
    "id": "ticket_001",
    "status": "CLOSED",
    "closed_at": "2026-10-08T16:00:00+05:30",
    "version": 4,
    "audit_id": "audit_008"
  }
}
```

Representative failures: 409 STATUS_TRANSITION_INVALID; 409 VERSION_CONFLICT. Common authorization and validation failures also apply.

Garage-owner performance statistics and formulas

Stories: AUS-21, AUS-22, AUS-23, AUS-24, AUS-25, AUS-26, AUS-27, AUS-28

• All statistics use the same normalized date/city/garage/service filters, permission scope and snapshot; exports include filters, generated timestamp and source totals.

• Jobs-created/status-cohort basis: jobs created in the selected period; status counts are their current status as_of. Status totals must reconcile to that filtered cohort job list. Do not mix created-in-period and completed-in-period counts without labels.

• Average jobs per garage = total cohort jobs / eligible garage count. Eligible garages include zero-job garages in the permitted segment; exclude inactive only when exclude_inactive=true. Return null, not zero, when denominator is zero.

• Conversion = converted_by_as_of / eligible_accepted_bookings * 100. Eligible cohort: accepted_at in the selected window, including bookings later CONVERTED; exclude invalid/deleted records under an approved policy. Numerator is the converted subset, not all converted bookings across unrelated dates.

• Average completion duration = sum(completed_at - created_at) / completed valid jobs in the created-in-period cohort, measured in hours. Exclude unfinished/cancelled/invalid-timestamp records and return exclusions/sample size; no samples means null.

• Average invoice amount = sum(final nonvoid invoice total_amount issued in period) / invoice count. Keep this separate from net collections and paid-invoice count. Collections use captured_at and refunds settled_at.

• Owner rankings use a deterministic tie-breaker (owner ID) and an explicit sort metric. Multi-garage owners are one aggregate row with garage_ids. Drill-down rows preserve owner identity and filter scope.

ADM-56 | Compare owners, rankings and operational/finance metrics

GET /api/v1/admin/garage-owner-statistics

Permission / proof: owner_statistics.read + finance.read for monetary fields | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&sort=jobs_created_desc&exclude_inactive=true&page=1&page_size=20

Field validation / operation rules: Optional owner_id and snapshot_id. Without finance.read omit monetary fields with restricted_fields metadata; do not return misleading zeroes.

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Compare owners, rankings and operational/finance metrics successful",
  "data": {
    "filters": {
      "date_from": "2026-10-01",
      "date_to": "2026-10-08",
      "timezone": "Asia/Kolkata",
      "city": "Hyderabad",
      "garage_id": null,
      "service_id": null
    },
    "snapshot_id": "snapshot_001",
    "generated_at": "2026-10-08T15:00:00+05:30",
    "items": [
      {
        "owner_id": "owner_001",
        "owner_name": "Example Owner A",
        "garage_ids": [
          "garage_001"
        ],
        "rank": 1,
        "jobs_created": 30,
        "status_counts": {
          "OPEN": 2,
          "IN_PROGRESS": 4,
          "WAITING_FOR_PARTS": 2,
          "QUALITY_CHECK": 1,
          "READY": 0,
          "COMPLETED": 20,
          "CANCELLED": 1
        },
        "eligible_accepted_bookings": 10,
        "converted_bookings": 8,
        "conversion_percent": 80,
        "completion_sample_size": 20,
        "average_completion_hours": 4.5,
        "overdue_jobs": 2,
        "gross_collected_amount": 2360000,
        "refunded_amount": 118000,
        "net_collected_amount": 2242000,
        "paid_invoice_count": 20,
        "issued_invoice_count": 20,
        "issued_invoice_total_amount": 2360000,
        "average_invoice_amount": 118000,
        "failed_attempts": 2,
        "currency": "INR"
      },
      {
        "owner_id": "owner_002",
        "owner_name": "Example Owner B",
        "garage_ids": [
          "garage_002"
        ],
        "rank": 2,
        "jobs_created": 20,
        "status_counts": {
          "OPEN": 1,
          "IN_PROGRESS": 3,
          "WAITING_FOR_PARTS": 2,
          "QUALITY_CHECK": 1,
          "READY": 1,
          "COMPLETED": 10,
          "CANCELLED": 2
```

```
        },
        "eligible_accepted_bookings": 10,
        "converted_bookings": 7,
        "conversion_percent": 70,
        "completion_sample_size": 10,
        "average_completion_hours": 6,
        "overdue_jobs": 3,
        "gross_collected_amount": 1180000,
        "refunded_amount": 0,
        "net_collected_amount": 1180000,
        "paid_invoice_count": 10,
        "issued_invoice_count": 10,
        "issued_invoice_total_amount": 1180000,
        "average_invoice_amount": 118000,
        "failed_attempts": 1,
        "currency": "INR"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 2,
    "source_totals": {
      "jobs_created": 50,
      "status_counts": {
        "OPEN": 3,
        "IN_PROGRESS": 7,
        "WAITING_FOR_PARTS": 4,
        "QUALITY_CHECK": 2,
        "READY": 1,
        "COMPLETED": 30,
        "CANCELLED": 3
      },
      "eligible_accepted_bookings": 20,
      "converted_bookings": 15,
      "net_collected_amount": 3422000,
      "paid_invoice_count": 30
    }
  }
}
```


Representative failures: 422 DATE_RANGE_INVALID; 409 SNAPSHOT_EXPIRED. Common authorization and validation failures also apply.

ADM-57 | Read reconcilable owner-performance totals/averages

GET /api/v1/admin/garage-owner-statistics/summary

Permission / proof: owner_statistics.read + finance.read for monetary fields | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&exclude_inactive=true

Field validation / operation rules: Return formulas/numerator/denominator and exclusions in metric metadata. Weighted completion mean: (20*4.5+10*6)/30=5 hours.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Read reconcilable owner-performance totals/averages successful",
  "data": {
    "filters": {
      "date_from": "2026-10-01",
      "date_to": "2026-10-08",
      "timezone": "Asia/Kolkata",
      "city": "Hyderabad",
      "garage_id": null,
      "service_id": null
    },
    "snapshot_id": "snapshot_001",
    "generated_at": "2026-10-08T15:00:00+05:30",
    "total_jobs_created": 50,
    "eligible_garages": 2,
    "average_jobs_per_garage": 25,
    "exclude_inactive": true,
    "status_counts": {
      "OPEN": 3,
      "IN_PROGRESS": 7,
      "WAITING_FOR_PARTS": 4,
      "QUALITY_CHECK": 2,
      "READY": 1,
      "COMPLETED": 30,
      "CANCELLED": 3
    },
    "eligible_accepted_bookings": 20,
    "converted_bookings": 15,
    "conversion_percent": 75,
    "completion_sample_size": 30,
    "completion_total_hours": 150,
    "average_completion_hours": 5,
    "net_collected_amount": 3422000,
    "paid_invoice_count": 30,
    "average_invoice_amount": 118000,
    "currency": "INR"
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-58 | Inspect one owner's performance and garage breakdown

GET /api/v1/admin/garage-owner-statistics/{owner_id}

Permission / proof: owner_statistics.read | Success HTTP: 200

Sample query: ?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad

Field validation / operation rules: Financial metrics may be included only with finance.read. Garage/job drill-down uses matching owner_id/garage_id filters.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Inspect one owner's performance and garage breakdown successful",
  "data": {
    "filters": {
      "date_from": "2026-10-01",
      "date_to": "2026-10-08",
      "timezone": "Asia/Kolkata",
      "city": "Hyderabad",
      "garage_id": null,
      "service_id": null
    },
    "snapshot_id": "snapshot_001",
    "generated_at": "2026-10-08T15:00:00+05:30",
    "owner_id": "owner_001",
    "jobs_created": 30,
    "completed_jobs": 20,
    "eligible_accepted_bookings": 10,
    "converted_bookings": 8,
    "conversion_percent": 80,
    "average_completion_hours": 4.5,
    "completion_sample_size": 20,
    "garage_breakdown": [
      {
        "garage_id": "garage_001",
        "jobs_created": 30,
        "overdue_jobs": 2
      }
    ]
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-59 | Read owner job/conversion/collection trends

GET /api/v1/admin/garage-owner-statistics/{owner_id}/trends

Permission / proof: owner_statistics.read | Success HTTP: 200

Sample query: ?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&period=daily

Field validation / operation rules: Provide explicitly labelled bucket date bases and numerators/denominators. Optional finance series requires finance.read.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Read owner job/conversion/collection trends successful",
  "data": {
    "filters": {
      "date_from": "2026-10-01",
      "date_to": "2026-10-08",
      "timezone": "Asia/Kolkata",
      "city": "Hyderabad",
      "garage_id": null,
      "service_id": null
    },
    "snapshot_id": "snapshot_001",
    "generated_at": "2026-10-08T15:00:00+05:30",
    "owner_id": "owner_001",
    "period": "daily",
    "series": [
      {
        "date": "2026-10-08",
        "jobs_created": 3,
        "jobs_completed": 2,
        "eligible_accepted_bookings": 2,
        "converted_bookings": 1,
        "conversion_percent": 50
      }
    ]
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

Operational, financial and owner-performance exports

Stories: AUS-18, AUS-28

• Exports are protected asynchronous resources, not public URLs. Require reports.export plus every data permission needed by the selected report; recheck at download.

• Supported proposed types: OPERATIONS, FINANCE, GARAGE_OWNER_PERFORMANCE. CSV and XLSX formats; sanitize spreadsheet-formula injection, redact restricted PII and define retention/size limits.

• Every export includes normalized filters, snapshot/as_of, generated timestamp, row count, metric definitions and source totals. GARAGE_OWNER_PERFORMANCE is one row per owner; multiple garage IDs are explicit.

• Status: QUEUED, PROCESSING, READY, FAILED, EXPIRED. Failed exports expose a safe error; never silently deliver partial results as complete. Capture format/requester/scope in audit.

ADM-60 | Request authorized filtered report

POST /api/v1/admin/reports/exports

Permission / proof: reports.export + selected report read permissions | Success HTTP: 202

Required header: Idempotency-Key: <unique_key>

Field validation / operation rules: report_type, format and filters required. snapshot_id optional; generate a fresh snapshot if omitted. exclude_inactive applies only to owner performance.

Request payload:

```
{
  "report_type": "GARAGE_OWNER_PERFORMANCE",
  "format": "CSV",
  "filters": {
    "date_from": "2026-10-01",
    "date_to": "2026-10-08",
    "timezone": "Asia/Kolkata",
    "city": "Hyderabad",
    "garage_id": null,
    "service_id": null
  },
  "snapshot_id": "snapshot_001",
  "exclude_inactive": true
}
```

Sample success response:

```
{
  "success": true,
  "message": "Request authorized filtered report successful",
  "data": {
    "id": "export_001",
    "status": "QUEUED",
    "report_type": "GARAGE_OWNER_PERFORMANCE",
    "format": "CSV",
    "audit_id": "audit_009"
  }
}
```

Representative failures: 403 REPORT_SCOPE_DENIED; 409 SNAPSHOT_EXPIRED; 413 EXPORT_TOO_LARGE; 429 EXPORT_LIMIT_REACHED. Common authorization and validation failures also apply.

ADM-61 | Poll report generation and reconcile totals

GET /api/v1/admin/reports/exports/{export_id}

Permission / proof: reports.export + selected report read permissions | Success HTTP: 200

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Poll report generation and reconcile totals successful",
  "data": {
    "id": "export_001",
    "status": "READY",
    "report_type": "GARAGE_OWNER_PERFORMANCE",
    "format": "CSV",
    "filters": {
      "date_from": "2026-10-01",
      "date_to": "2026-10-08",
      "timezone": "Asia/Kolkata",
      "city": "Hyderabad",
      "garage_id": null,
      "service_id": null
    },
    "snapshot_id": "snapshot_001",
    "generated_at": "2026-10-08T15:05:00+05:30",
    "row_count": 2,
    "source_totals": {
      "jobs_created": 50,
      "net_collected_amount": 3422000
    },
    "download_path": "/api/v1/admin/reports/exports/export_001/download",
    "expires_at": "2026-10-09T15:05:00+05:30"
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-62 | Download protected report

GET /api/v1/admin/reports/exports/{export_id}/download

Permission / proof: reports.export + selected report read permissions | Success HTTP: 200

Field validation / operation rules: Requester or approved supervisory scope only; verify current permission, READY status and expiry. XLSX uses its correct MIME type.

Request payload:

```
No request body.
```

Sample success response:

```
Binary response: Content-Type: text/csv; Content-Disposition: attachment;
  filename="garage-owner-performance.csv"
```

Representative failures: 409 EXPORT_NOT_READY; 410 EXPORT_EXPIRED. Common authorization and validation failures also apply.

Audit trail and unified customer/garage activity

Stories: AUS-19, AUS-29

• Audit records cover successful/failed login, status/account/role changes, financial actions, announcements and exports; actor, time, IP/device, result, correlation ID and approved before/after details are retained.

• Audit is append-only; no admin update/delete audit operation. Protect logs from secret/OTP/token/raw gateway payload leakage. IP/device data is privileged, masked or policy-limited.

• Unified activity links customer booking, garage acceptance, job creation/status, invoice, payment and review events using stable IDs/correlation. Missing events are shown explicitly, not invented.

• Financial activity fields require finance.read. Audit log read requires audit.read and is itself auditable.

ADM-63 | Search audited administrative/security actions

GET /api/v1/admin/audit-logs

Permission / proof: audit.read | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&actor_id=admin_001&action=GARAGE_SUSPENDED&result=SUCCESS&page=1&page_size=20

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Search audited administrative/security actions successful",
  "data": {
    "items": [
      {
        "id": "audit_002",
        "actor_id": "admin_001",
        "action": "GARAGE_SUSPENDED",
        "resource": {
          "type": "GARAGE",
          "id": "garage_002"
        },
        "timestamp": "2026-10-08T14:00:00+05:30",
        "ip_address": "192.0.2.10",
        "device": "Example workstation",
        "result": "SUCCESS",
        "reason": "Partner quality review",
        "correlation_id": "correlation_001"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 1
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-64 | Inspect safe audit details

GET /api/v1/admin/audit-logs/{audit_id}

Permission / proof: audit.read | Success HTTP: 200

Request payload:

```
No request body.
```

Sample success response:

```
{
  "success": true,
  "message": "Inspect safe audit details successful",
  "data": {
    "id": "audit_002",
    "actor_id": "admin_001",
    "action": "GARAGE_SUSPENDED",
    "resource": {
      "type": "GARAGE",
      "id": "garage_002"
    },
    "before": {
      "operational_status": "ACTIVE"
    },
    "after": {
      "operational_status": "SUSPENDED"
    },
    "result": "SUCCESS",
    "reason": "Partner quality review",
    "timestamp": "2026-10-08T14:00:00+05:30"
  }
}
```

Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-65 | Monitor linked customer and garage service activity

GET /api/v1/admin/activity

Permission / proof: activity.read | Success HTTP: 200

Sample query:
?date_from=2026-10-01&date_to=2026-10-08&timezone=Asia%2FKolkata&city=Hyderabad&customer_id=customer_001&garage_id=garage_001&page=1&page_size=20

Field validation / operation rules: Optional owner_id, event_type, booking_id, job_id, invoice_id filters; apply linked-domain permissions.

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Monitor linked customer and garage service activity successful",
  "data": {
    "items": [
      {
        "id": "activity_001",
        "event_type": "BOOKING_CREATED",
        "customer_id": "customer_001",
        "owner_id": "owner_001",
        "garage_id": "garage_001",
        "booking_id": "booking_001",
        "timestamp": "2026-10-07T15:00:00+05:30"
      },
      {
        "id": "activity_002",
        "event_type": "JOB_CREATED",
        "customer_id": "customer_001",
        "owner_id": "owner_001",
        "garage_id": "garage_001",
        "booking_id": "booking_001",
        "job_id": "job_001",
        "timestamp": "2026-10-08T10:00:00+05:30"
      }
    ],
    "page": 1,
    "page_size": 20,
    "total": 2
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

ADM-66 | Inspect complete linked service funnel

GET /api/v1/admin/activity/service-funnels/{booking_id}

Permission / proof: activity.read + relevant linked-record permissions | Success HTTP: 200

Request payload:

```
No request body.
```

Sample success response:


```
{
  "success": true,
  "message": "Inspect complete linked service funnel successful",
  "data": {
    "booking_id": "booking_001",
    "customer_id": "customer_001",
    "garage_id": "garage_001",
    "job_id": "job_001",
    "invoice_id": "invoice_001",
    "events": [
      {
        "type": "BOOKING_CREATED",
        "timestamp": "2026-10-07T15:00:00+05:30"
      },
      {
        "type": "BOOKING_ACCEPTED",
        "timestamp": "2026-10-07T16:00:00+05:30"
      },
      {
        "type": "JOB_CREATED",
        "timestamp": "2026-10-08T10:00:00+05:30"
      },
      {
        "type": "JOB_COMPLETED",
        "timestamp": "2026-10-08T12:00:00+05:30"
      },
      {
        "type": "INVOICE_PUBLISHED",
        "timestamp": "2026-10-08T12:10:00+05:30"
      },
      {
        "type": "PAYMENT_CONFIRMED",
        "payment_id": "payment_001",
        "timestamp": "2026-10-08T12:15:00+05:30"
      },
      {
        "type": "REVIEW_SUBMITTED",
        "review_id": "review_001",
        "timestamp": "2026-10-08T12:20:00+05:30"
      }
    ]
  }
}
```


Representative failures: 404 RESOURCE_NOT_FOUND. Common authorization and validation failures also apply.

4. Existing baseline and requirements gaps

The observed admin router provides menus/identity, dashboard, owner-oriented users/status, garages/detail, jobs/detail, booking list, basic analytics, billing-issue review and audit list. These are a starting point, not complete acceptance of the CSV.

Area	Observed baseline	Required gap / proposed extension
Secure access	Dedicated protected admin identity/session mechanism exists.	MFA/OTP requirement must be verified/added; private routes are withheld.
Dashboard/filtering	Dashboard/basic analytics routes exist.	Full KPI set, city/service/date consistency, snapshots and explicit metric bases.
Partners/customers	Owner/garage reads; user status update is owner-oriented.	Owner aggregates, separate customer identity views, garage activation/verification control, risk/activity fields.
Bookings/jobs	Booking list and job list/detail exist.	Decision reasons, timeline/exception trends, delay thresholds/alerts and consistent drill-down.
Finance	Read-only billing issue review exists.	Confirmed invoices/payments/refunds/disputes, safe gateway evidence and reconciled financial trends.
Communications/support	Not certified by this router review.	Delivery monitoring, announcement audience/send/results and support assignment/replies/closure.
RBAC/audit	Admin protection/basic audit list exists.	Fine-grained roles/scopes, permission-aware exports, full actor/result/device/action coverage.
Owner performance/exports/activity	Not certified by this router review.	AUS-21..29 formulas, reconciled cohort statistics, owner exports and linked service funnel.

All contracts in this approval document are proposed requirements, including changes to existing business paths. Existing response shapes may differ. This is a limited baseline review, not certification of every database/service capability or migration plan.

5. Story-to-API traceability

Story	Source epic	Priority	Covering operations
AUS-1	Admin Authentication	Must Have	ADM-01, ADM-02, ADM-03, ADM-04
AUS-2	Executive Dashboard	Must Have	ADM-11, ADM-12
AUS-3	Executive Dashboard	Must Have	ADM-11, ADM-12
AUS-4	Garage Operations	Must Have	ADM-13, ADM-14, ADM-15, ADM-16, ADM-17, ADM-18, ADM-19
AUS-5	Garage Operations	Must Have	ADM-13, ADM-14, ADM-15, ADM-16, ADM-17, ADM-18, ADM-19
AUS-6	Garage Operations	Must Have	ADM-13, ADM-14, ADM-15, ADM-16, ADM-17, ADM-18, ADM-19
AUS-7	Customer Operations	Must Have	ADM-20, ADM-21, ADM-22, ADM-23
AUS-8	Customer Operations	Should Have	ADM-20, ADM-21, ADM-22, ADM-23
AUS-9	Booking Monitoring	Must Have	ADM-24, ADM-25, ADM-26
AUS-10	Booking Monitoring	Should Have	ADM-24, ADM-25, ADM-26
AUS-11	Job Monitoring	Must Have	ADM-27, ADM-28, ADM-29, ADM-30, ADM-31
AUS-12	Job Monitoring	Must Have	ADM-27, ADM-28, ADM-29, ADM-30, ADM-31
AUS-13	Finance	Must Have	ADM-32, ADM-33, ADM-34, ADM-35, ADM-36, ADM-37, ADM-38, ADM-39, ADM-40, ADM-41
AUS-14	Finance	Must Have	ADM-32, ADM-33, ADM-34, ADM-35, ADM-36, ADM-37, ADM-38, ADM-39, ADM-40, ADM-41
AUS-15	Notifications	Must Have	ADM-42, ADM-43, ADM-44, ADM-45, ADM-46, ADM-47, ADM-48
AUS-16	Notifications	Should Have	ADM-42, ADM-43, ADM-44, ADM-45, ADM-46, ADM-47, ADM-48
AUS-17	Support	Must Have	ADM-49, ADM-50, ADM-51, ADM-52, ADM-53, ADM-54, ADM-55
AUS-18	Reports	Should Have	ADM-60, ADM-61, ADM-62
AUS-19	Audit and Security	Must Have	ADM-63, ADM-64, ADM-65, ADM-66
AUS-20	Audit and Security	Must Have	ADM-05, ADM-06, ADM-07, ADM-08, ADM-09, ADM-10
AUS-21	Garage Owner Statistics	Must Have	ADM-56, ADM-57, ADM-58, ADM-59
AUS-22	Garage Owner Statistics	Must Have	ADM-56, ADM-57, ADM-58, ADM-59
AUS-23	Garage Owner Statistics	Should Have	ADM-56, ADM-57, ADM-58, ADM-59
AUS-24	Garage Owner Statistics	Must Have	ADM-56, ADM-57, ADM-58, ADM-59
AUS-25	Garage Owner Statistics	Must Have	ADM-56, ADM-57, ADM-58, ADM-59
AUS-26	Garage Owner Statistics	Must Have	ADM-56, ADM-57, ADM-58, ADM-59
AUS-27	Garage Owner Statistics	Must Have	ADM-56, ADM-57, ADM-58, ADM-59
AUS-28	Garage Owner Statistics	Should Have	ADM-56, ADM-57, ADM-58, ADM-59, ADM-60, ADM-61, ADM-62
AUS-29	Customer and Garage Activity	Must Have	ADM-63, ADM-64, ADM-65, ADM-66

AUS-1 is covered by logical private authentication operations without exposing URLs. AUS-21..28 use the performance contracts and formulas; AUS-18/28 share authorized export operations. All 29 source stories are mapped.

6. Client decision register

ID	Topic	Approval required
AD-01	Admin security	Approve MFA method, factor enrollment/recovery, session expiry, recent-MFA checks, trusted-device policy and private route distribution.
AD-02	Roles and scope	Approve CEO/operations/finance/support roles, granular permissions, global/garage scopes and authority for role assignment.
AD-03	Partner status	Approve activation/suspension/verification rules, required evidence and handling of existing jobs/bookings during suspension.
AD-04	Customer monitoring	Approve inactivity threshold, blocked-state source and privacy/masking. Customer blocking/impersonation is not included.
AD-05	Metric cohorts	Approve event date bases, accepted-booking conversion cohort, job status cohort, zero-activity garage inclusion and snapshot behavior.
AD-06	Financial definitions	Approve net collections versus accounting revenue, refund/tax/discount treatment, invoice averages and provider dispute evidence. No new financial writes.
AD-07	Delay thresholds	Approve state/service thresholds, due dates, alert severity/deduplication and intervention ownership.
AD-08	Announcements	Approve channels, consent/purpose, audience maximum, previews, recipient suppression and delivery/retry rules.
AD-09	Support	Approve categories, priorities, transitions, assignment eligibility, external replies, closure/reopening and SLA.
AD-10	Exports	Approve CSV/XLSX formats, data masking, report row/range limits, retention/expiry and supervisory download access.
AD-11	Audit/privacy	Approve logged security/action fields, IP/device access, retention, redaction and evidence controls.

Proposed defaults are not approved production settings. Resolve these decisions before final API implementation or estimating delivery.